香 港 大 學

THE UNIVERSITY OF HONG KONG

COMP7705

MSc(CompSc) Project

Detailed Project Proposal

An innovative AI-powered interview assessment tool for company

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Abstract

The global AI recruitment market is growing rapidly, with companies like Amazon using AI for 60-70% of initial applicant interactions. "Candidates who are identified with AI job matching technology and reach the interview stage are 24% more likely to receive a positive outcome after their initial interview loop" (**Parulekar, 2026**). In Hong Kong, AI adoption in HR is accelerating amid talent shortage.

In this study, we aim to develop an innovative AI-powered virtual interviewer tailored for HR teams and hiring managers to streamline initial candidate screening for technical and behavioral roles.

We will use Python, web interface, and AI API as the core LLM engine. The outcome will demonstrate benefits (efficiency, consistency) and address challenges (bias, privacy) in AI recruitment.

Objective

The primary objectives are to:

- Develop a basic AI tool that automates resume screening by matching candidate qualifications to job descriptions, reducing human effort
- Generate customized interview questions based on job types
- Analyze candidate responses for relevance and quality, providing a ranking score to shortlist top applicants
- Demonstrate how AI can minimize bias and improve efficiency in hiring, while learning practical skills in Python, web apps, and data processing.

Brief Literature Review

Importance of AI in Recruitment:

AI tools are transforming hiring. **Muhaiminul Islam, Md. Mahbubur Rahman, Md. Abu Taher, G. M. Azmal Ali Quaosar & Md. Aftab Uddin (2024)** shows AI can simplify and streamline the hiring process. Nearly all Fortune 500 companies use ATS and AI for massive volumes. Amazon example: 1.5M+ workforce, 60–70% applicants first meet AI.

Popular Platforms & Examples:

TalentLy.ai (Asia's first AI interactive interview platform) conducts live interviews, analyzes responses using NLP and LLMs. Benefits include faster screening; challenges include potential bias and lack of transparency. Hirevue is founded in 2004 and one of the largest AI recruitment platforms worldwide used in over 180 countries with millions of candidates assessed.

Benefits and Challenges:

An innovative AI-powered interview assessment tool conducts real-time multi-turn interviews, automatically scores responses on relevance, technical depth, communication clarity, and value alignment, and provides intelligent follow-up probes with comprehensive reports and hiring recommendations. Key benefits include 50–80% reduction in HR screening time, enhanced consistency and objectivity to minimize human bias, improved evaluation depth through adaptive questioning, and high scalability with low marginal cost

On the other hand, algorithmic bias in screening is one of the challenges in AI hiring. "One of the best-known real-world examples is the case of Amazon in 2018, where a tested AI software systematically discriminated against women in the hiring process." **(Hunkenschroer and Kriebitz, 2023)**

Technology Behind

The technology stack for this AI-powered virtual interviewer is designed as below. Python serves as the primary programming language. Streamlit is chosen as the web framework to create an interactive chat interface. The core conversational and scoring logic relies on Large Language Models accessed via OpenAI API (GPT-4o or GPT-3.5-turbo) or Grok API for fast and cost-effective inference. Retrieval-Augmented Generation (RAG) is implemented simply using company values and job descriptions as context in prompts. For future extensions, Whisper (open-source) can handle speech-to-text, OpenCV + MediaPipe for basic facial emotion analysis, and SQLite for storing interview logs and reports. This modular, open-source approach ensures independent learning, low cost, and alignment with the project's emphasis on self-motivation and resourcefulness.

Milestones

Tasks	Estimated completion time	Estimated number of learning hours
1. Research & Planning - Review AI Hiring PDF and related literature - Define objectives, scope, and stream (Multimedia Computing) - Install Python, Streamlit, OpenAI/Groq API - Draft and submit Detailed Project Proposal - Create and initialize project webpage on school account	March 10, 2026	60 hours

Tasks	Estimated completion time	Estimated number of learning hours
2. Basic Interface & Question Generation - Build simple chat interface using Streamlit - Implement basic question generation with LLM API (prompt engineering) - Test with sample job descriptions - Update project webpage with screenshots and progress	April 7, 2026	50 hours
3. Multi-turn Conversation & Initial Scoring - Add conversation history management for multi-turn interviews - Develop initial automated scoring (relevance, technical depth) - Conduct preliminary tests with dummy responses - Update webpage with demo video/screenshots	May 5, 2026 (Progress Update 2)	45 hours
4. Advanced Features & Interim Report - Implement full scoring dimensions (clarity, value alignment) - Add intelligent follow-up probes - Generate basic reports and recommendations - Prepare and deliver Interim Report & Presentation	June 1, 2026 (Interim Report & Presentation)	50 hours

Tasks	Estimated completion time	Estimated number of learning hours
5. Testing, Refinement & Bias Check - Full system end-to-end testing with 20–30 dummy cases - Check for bias and improve prompts - Fix bugs and optimize performance - Update webpage with test results	June 16, 2026 (Progress Update 3)	70 hours
6. Final Polish & Oral Preparation - Refine UI and report format - Conduct final full tests and collect results - Prepare demo video and oral examination materials - Final webpage update	July 6, 2026 (Progress Update 4)	50 hours
7. Final Submission - Complete Project Report - Submit final webpage (July 13)	July 17, 2026 (Project Report)	30 hours

Deliverables

1. Research Proposal (this document)
2. Milestone & Progress Reports
3. Testing Reports (Screening, LLM Analysis, Bias Report)
4. Improvement Recommendations & Final Report (July 17, 2026)
5. Presentation & Supporting Materials (project webpage + demo)
6. Project Webpage (updated throughout, Guideline Page 3)

Reference

Parulekar, A. (2026, January 22). *How 3 AI innovations are creating a better experience for candidates in Amazon's hiring process.* Amazon News. https://www.aboutamazon.com/news/workplace/artificial-intelligence-resume-jobs-hiring-amazon

Islam, M., Rahman, M. M., Taher, M. A., Quoasar, G. M. A. A., and Uddin, M. A. (2024, January 22). *Using artificial intelligence for hiring talents in a moderated mechanism.* Springer Nature Link. https://link.springer.com/article/10.1186/s43093-024-00303-x

Hunkenschroer, A.L. and Kriebitz, A. (2022, July 25). *Is AI recruiting (un)ethical? A human rights perspective on the use of AI for hiring.* Springer Nature Link. https://link.springer.com/article/10.1007/s43681-022-00166-4